

**Department of Electrical and Electronics Engineering****Academic Year 2024 – 2025 (Odd Semester)****Degree, Semester & Branch:** VII Semester B.E. EEE**Course Code & Title:** EE3701 High Voltage Engineering**Name of the Faculty member (s):** Dr. S. Narthana**Innovative Practice Description**

- **Unit / Topic:** Unit I / Protection against over voltages
- **Course Outcome:** CO 1
- **Unit Outcome:** TLO6
- **Activity Chosen:** One Minute Paper
- **Justification:**

The identification of the protection methods against the different types of overvoltages are crucial for the effective utilization. A one-minute paper is used for this concept because it prompts students to quickly summarize key points and reflect on their understanding of overvoltage protection. This technique helps in identifying the appropriate methods and improves the students' learning with the focused thinking.

- **Time Allotted for the Activity:** 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.

- **CO – PO / PSO mapping:**

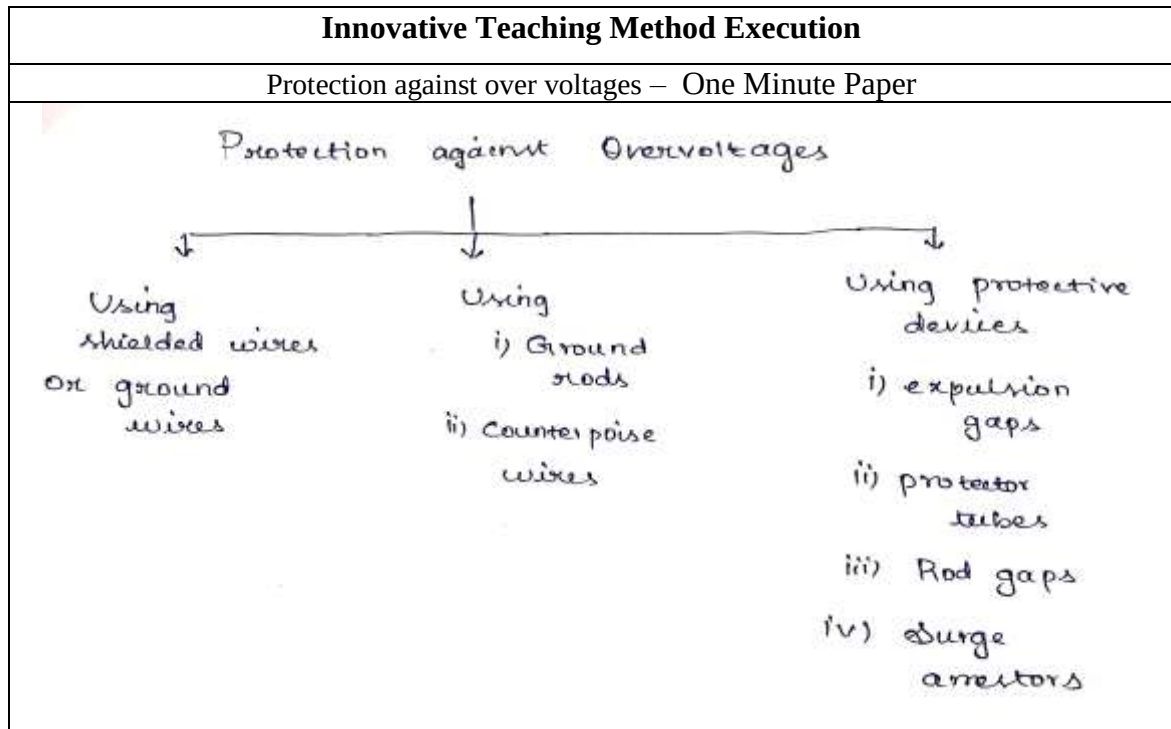
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	1	1	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9
	1
Justification for correlation	The students can function effectively as an individual

- Images / Screenshot of the practice:



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- ❖ **Benefit of the practice:**
- ❖ Students can be able to attend the question even the questions are in indirect form.
- ❖ Students can able to explain the concepts in examination without any confusion.
- ❖ **Challenges faced in implementation:**
- 1. Time utilization for conducting activity.

References:

1. S.Naidu and V. Kamaraju, 'High Voltage Engineering', Tata McGraw Hill, Fifth Edition, 2013.
2. E. Kuffel and W.S. Zaengl, J.Kuffel, 'High voltage Engineering fundamentals', Newness Second Edition, Elsevier, New Delhi, 2005.
3. C.L. Wadhwa, 'High voltage Engineering', New Age International Publishers, Fourth Edition, 2020.

Signature of Faculty Member

HOD

**Department of Electrical and Electronics Engineering****Academic Year 2024 – 2025 (Odd Semester)****Degree, Semester & Branch:** VII Semester B.E. EEE**Course Code & Title:** EE3701 High Voltage Engineering**Name of the Faculty member (s):** Dr. S. Narthana**Innovative Practice Description**

- **Unit / Topic:** Unit II / Applications of insulating materials in electrical equipments
- **Course Outcome:** CO 2
- **Unit Outcome:** TLO13
- **Activity Chosen:** Class poll
- **Justification:**

This topic is chosen, since polls can help students feel valued and make them think about the concept. They can also help the lecture more informative and make it interesting. Students can improve their understanding on this topic by this activity. More time for discussion is given the class unlike traditional learning. This activity greatly helps the student to know the concept much deeper.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

The students are separated into teams and were asked to work as a team to answer the questions that I asked during this active learning approach. They should discuss among themselves and one among them was asked to present the answers.

- The class poll questions have been shown to the class using the classroom projector.
- The teams have provided their choices of the answers along with their reasons.
- Finally, the correct and explanation have been provided by me.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	2	-	-	1	-	-	1	-	-	-	1	1	-	-

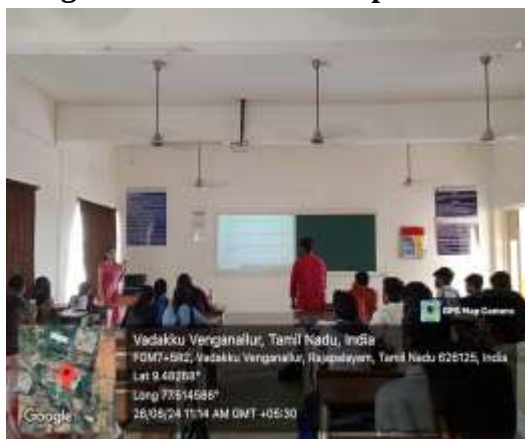
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10
	1	1

Justification for correlation	The students are working in a small group while doing this activity. Hence it is mapped with level 1.	The students are communicating with each other in a team for getting the answer for the poll. Hence their communication skills are getting improved. Hence it is mapped with level 1.
--------------------------------------	---	---

• **Images / Screenshot of the practice:**



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Students understood the concept which was reflected from their answers for the questions I have asked during class poll session.
- ❖ **Benefit of the practice:**
- ❖ Students can be able to attend and answer the question even the questions are in indirect form.
- ❖ Students can be able to explain the concepts in examination without any confusion.
- ❖ **Challenges faced in implementation:**
- 1. Time utilization for conducting activity.

References:

1. S.Naidu and V. Kamaraju, 'High Voltage Engineering', Tata McGraw Hill, Fifth Edition, 2013.
2. E. Kuffel and W.S. Zaengl, J.Kuffel, 'High voltage Engineering fundamentals', Newness Second Edition, Elsevier, New Delhi, 2005.

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE

Course Code & Title: EE3701 High Voltage Engineering

Name of the Faculty member (s): Dr. S. Narthana

Innovative Practice Description

- **Unit / Topic:** Unit III / High voltage and current generation and Sphere gaps
- **Course Outcome:** CO 3
- **Unit Outcome:** TLO14, TLO16, TLO22
- **Activity Chosen:** Virtual Laboratory
- **Justification:**

The method and procedure of the high voltage and current generation, sphere gap measurement is taught using this active learning approach of virtual laboratory. This lab helps students to apply the theoretical concepts virtually to prove themselves what they have learned through reading or didactic teaching. This approach can strengthen students' logical thinking processes and test their mental model of a process. The activity can be done individually or in pairs.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

This innovative practice is conducted for IV year EEE students, after explained the concept of high voltage and current generation and sphere gap measurement techniques.

- The students are taken to the Power system simulation laboratory and each one is occupying the one system with Internet connectivity.
- The link for the virtual laboratory is shown to them through the LCD projector and I told them to go that page.
- I explained about the virtual experiments and how to do those experiments step by step. I also told my students to do the same parallel.
- I changed few parameters in those experiments and shown the waveform changes. I also explained the reasons for those changes.
- Then I made them to discuss about the concepts that was taught in the class for 10 minutes and asked the students to start the experiment and take the corresponding results for 20 minutes.
- Finally, the students have taken the screenshots of their laboratory experiment setup and resultant waveform and submitted as an Assignment.

- **CO – PO / PSO mapping:**

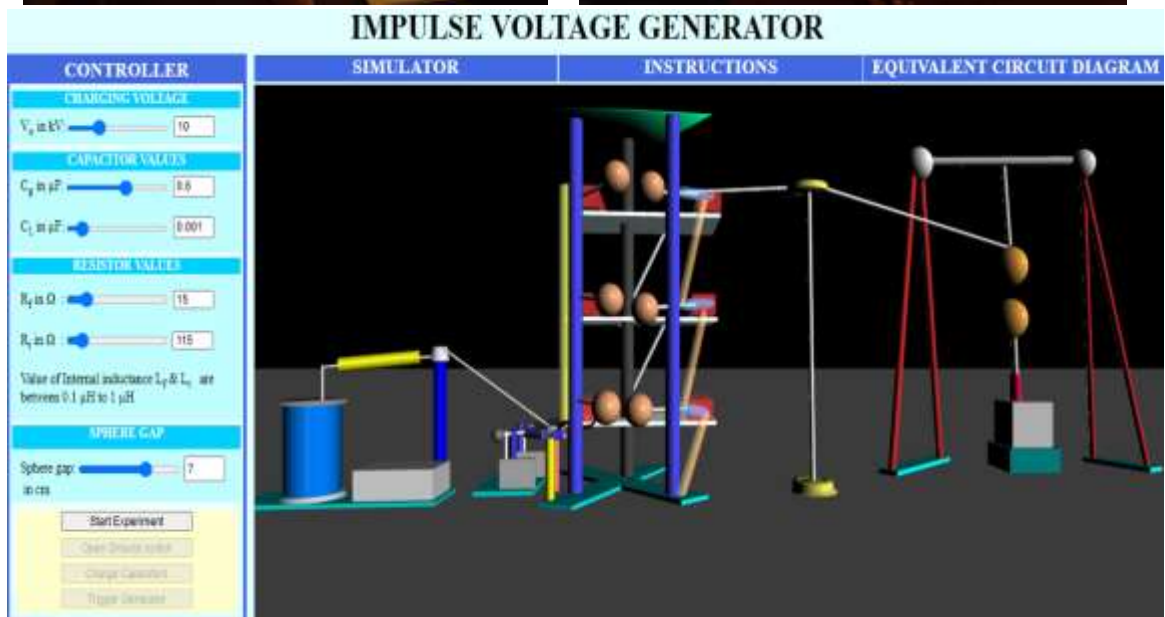
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	-	-	1	-	-	1	-	-	-	1	1	-	1

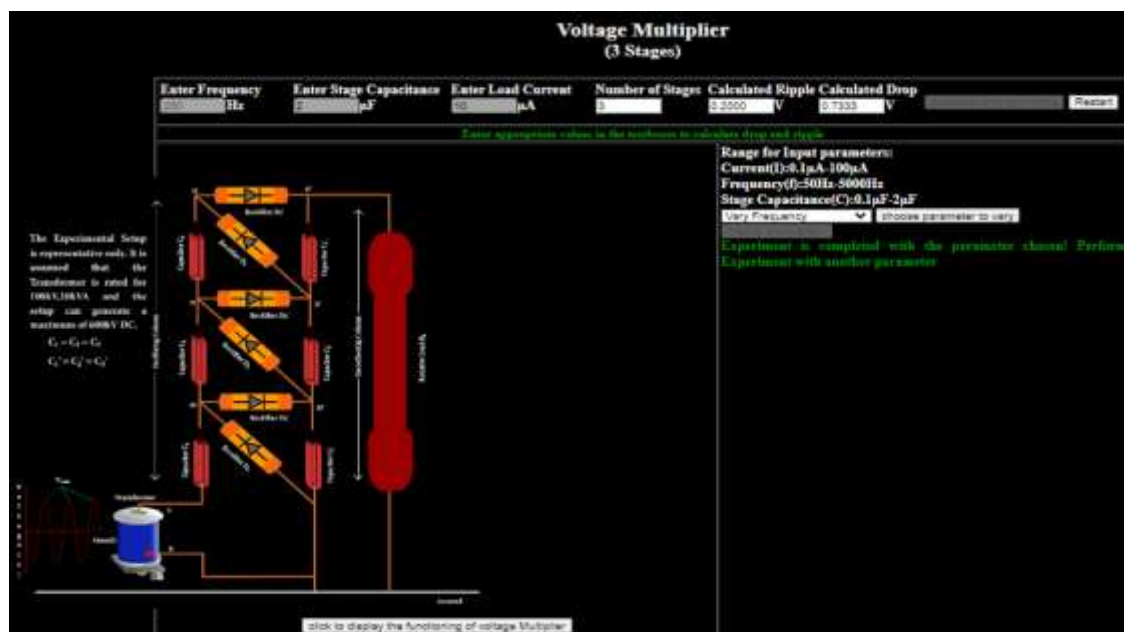
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10
	1	1
Justification for correlation	The students can function effectively as an individual.	The students are writing the report for the Virtual laboratory experiments. Hence their written communication skills are getting improved. Hence it is mapped with level 1.

- **Images / Screenshot of the practice:**





Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**
- ❖ Students understood the principle of working of the high voltage and current generation by this active learning approach and the students learnt the concepts interestingly.
- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
The students actively participated in the virtual lab practice. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to clear all the student doubt and make them more involved in the learning process.
- ❖ **Challenges faced in implementation:**
 - ✓ Few students had a problem to navigate through the virtual experiments; I helped them to do it.
 - ✓ Managing the students, making them to do the experiments and clearing the doubts about the tool and concepts took lot of time. Hence, it is not possible to cover all the experiments in one period.

References:

- ❖ S.Naidu and V. Kamaraju, 'High Voltage Engineering', Tata McGraw Hill, Fifth Edition, 2013.
- ❖ <http://vlabs.iitkgp.ernet.in/vhv/>

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE

Course Code & Title: EE3701 High Voltage Engineering

Name of the Faculty member (s): Dr. S. Narthana

Innovative Practice Description

- **Unit / Topic:** Unit IV / Insulation coordination

- **Course Outcome:** CO 4

- **Unit Outcome:** TLO28

- **Activity Chosen:** Ungraded Quiz

- **Justification:**

The chosen topic will have lot of statistical methods to decide the insulation level for UHV and EHV systems. Usually the students will find it difficult to understand the methods and also to properly select the insulation level for the given power apparatus. By conducting this ungraded quiz, I tried to help them to understand the concepts better. Improved the quality of student participation by recalling the important aspects about the terminologies involved in the insulation coordination of the various EHV and UHV equipment.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

Quiz activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. The quiz has been created in the Quiz online tool and pass code has been shared with the students.

- All the students have started the quiz at the same time. At the end of the quiz, the correct answers will be shown to them in the screen.
- Finally, I explained the correct answer for the quiz questions with proper reasons.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	-	-	1	-	-	-	-	-	-	1	-	-	1

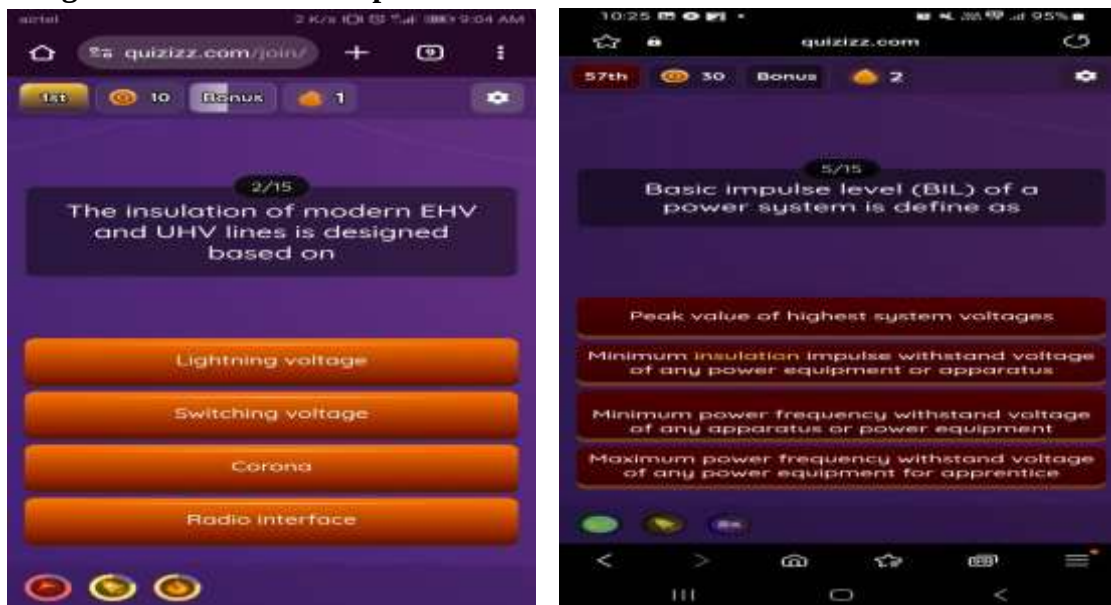
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 5	PO 8
	1	1

Justification for correlation	The students are using online tool for doing this quiz. Hence it is mapped with this PO at level 1	The students are doing this quiz without discussing and copying from others. Hence it is mapped with this PO at level 1
--------------------------------------	--	---

• **Images / Screenshot of the practice:**



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

Students understood the important terminologies and design principles present in this topic and improved the ability to relate their answers to the classroom learning.

❖ **Benefit of the practice:** The benefits of quiz are a great way for students to make notes on all of the information they receive. It made the students the general steps involved in the process and make the students to implement the procedure for the particular topic.

❖ **Challenges faced in implementation:**

Normally teachers will give longer explanations in the notes section of the topic. The students are made to answer the multiple-choice questions, fill ups for the topic of insulation coordination only for 15 mins. But in real scenario, it takes 25 mins.

References:

1. S.Naidu and V. Kamaraju, 'High Voltage Engineering', Tata McGraw Hill, Fifth Edition, 2013.
2. E. Kuffel and W.S. Zaengl, J.Kuffel, 'High voltage Engineering fundamentals', Newness Second Edition, Elsevier, New Delhi, 2005.
3. <https://site.ieee.org/sas-pesias/files/2020/02/InsulationCoordination.pdf>

Signature of Faculty Member

HOD

**Department of Electrical and Electronics Engineering****Academic Year 2024 – 2025 (Odd Semester)****Degree, Semester & Branch:** VII Semester B.E. EEE**Course Code & Title:** EE3701 High Voltage Engineering**Name of the Faculty member (s):** Dr. S. Narthana**Innovative Practice Description**

- **Unit / Topic:** Unit V / Manufacturing of sand paper, electrostatic pumping and propulsion
- **Course Outcome:** CO 5
- **Unit Outcome:** TLO32, TLO33
- **Activity Chosen:** Flipped Classroom

This topic is chosen, since it can be easily understood by students. Students can improve their understanding on this topic by this activity. More time for discussion is given in the class unlike traditional learning. This activity greatly helps the student to know the concept much deeper.

- **Time Allotted for the Activity:** 25 Minutes

- **Details of the Implementation:**

Each team was asked to do literature survey about the given topic. They should include at least 5 Reference material about the given topic for which the material is already shared. Students in a team were asked to present their topics. Queries will be asked by the students to the presenting team.

- **CO – PO / PSO mapping:**

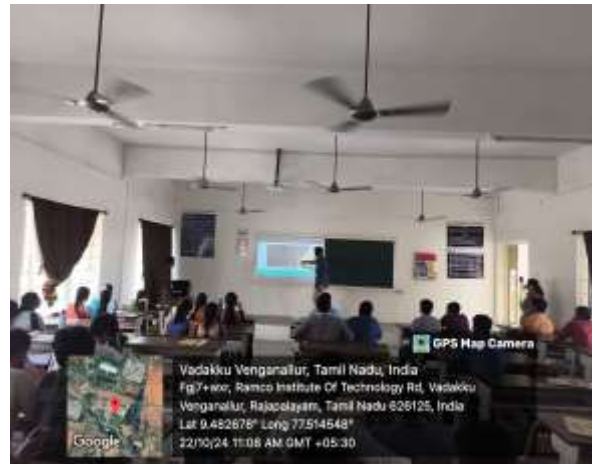
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	-	-	-	-	-	-	-	-	-	1	1	-	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 11
	1
Justification for correlation	The students are able to demonstrate the knowledge and understanding of the given topic and identify the tasks.

- **Images / Screenshot of the practice:**



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

The students enjoyed this Flipped class room activity.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students actively participated in the flipped classroom activity. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to clear all the student doubt and make them more involved in the learning process.

- ❖ **Challenges faced in implementation:**

Still few students were hesitant to convey their points and queries. Few students have not taken this activity seriously and remain quite in the class.

References:

1. Mazen Abdel – Salem, Hussein Anis, Ahdab A-Morshedy, Roshday Radwan, High Engineering – Theory & Practice, Second Edition, Taylor & Francis Group, 2019.

Signature of Faculty Member

HOD